Yakeen Plus (2024)

PRACTICE TEST - 06

DURATION : 200 Minutes | **DATE : 23/10/2024** | **M. MARKS : 720**

Topics Covered

Physics :	Oscillations, Waves, Electric Charges and Fields, Electrostatic Potential and Capacitance
Chemistry :	Organic Chemistry: Some Basic principles and Techniques (IUPAC Naming), Organic Chemistry: Some Basic principles and Techniques (Isomerism), Organic Chemistry: Some Basic principles and Techniques (GOC), Hydrocarbon
Biology :	**(Botany) :** Anatomy of Flowering Plants, Respiration in Plants **(Zoology) :** Neural Control & Coordination, Chemical Coordination & Integration

General Instructions:

1. Immediately fill in the particulars on this page of the test booklet.
2. The test is of **3 hours 20 min**. duration.
3. The test booklet consists of **200** questions. The maximum marks are **720**.
4. There are four Section in the Question Paper, Section I, II, III & IV consisting of Section-I **(Physics)**, Section-II **(Chemistry)**, Section-III **(Botany)** & Section IV **(Zoology)** and having **50 Questions** in each Subject and each subject is divided in two Section, **Section A** consisting 35 questions (all questions are compulsory) and **Section B** consisting 15 Questions (Any 10 questions are compulsory).
5. There is only one correct response for each question.
6. Each correct answer will give 4 marks while 1 Mark will be deducted for a wrong MCQ response.
7. No student is allowed to carry any textual material, printed or written, bits of papers, pager, mobile phone, any electronic device, etc. inside the examination room/hall.
8. On completion of the test, the candidate must hand over the Answer Sheet to the Invigilator on duty in the Room/Hall. However, the candidates are allowed to take away this Test Booklet with them.

OMR Instructions:

1. Use blue/black dark ballpoint pens.
2. Darken the bubbles completely. Don't put a tick mark or a cross mark where it is specified that you fill the bubbles completely. Half-filled or over-filled bubbles will not be read by the software.
3. Never use pencils to mark your answers.
4. Never use whiteners to rectify filling errors as they may disrupt the scanning and evaluation process.
5. Writing on the OMR Sheet is permitted on the specified area only and even small marks other than the specified area may create problems during the evaluation.
6. Multiple markings will be treated as invalid responses.
7. **Do not fold or make any stray mark on the Answer Sheet (OMR).**

Name of the Student (In CAPITALS) : ____________________

Roll Number : ____________________

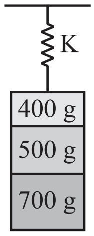
OMR Bar Code Number : ____________________

Candidate's Signature : ____________________ **Invigilator's Signature** ____________________

SECTION-(I) PHYSICS

SECTION - A

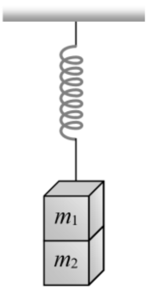
1. Three masses 700 g, 500 g and 400 g are suspended at the end of a spring as shown and are in equilibrium. When the 700 g mass is removed, the system oscillates with a period of 3 seconds, when the 500 gm mass is also removed, it will oscillate with a period of:



(1) 1 s (2) 2 s
(3) 3 s (4) $\sqrt{\frac{12}{5}}$ s

2. The second overtone of an open organ pipe has the same frequency as the first overtone of a closed pipe L metre long. The length of the open pipe will be:
(1) L (2) $2L$
(3) $\frac{L}{2}$ (4) $4L$

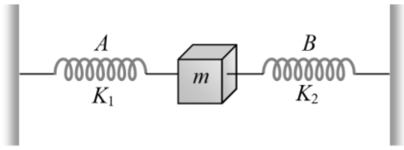
3. If a spring extends by x on loading, then energy stored by the spring is: (if T is the tension in the spring and K is the spring constant)



(1) $\frac{T^2}{2x}$ (2) $\frac{T^2}{2K}$
(3) $\frac{2K}{T^2}$ (4) $\frac{2T^2}{K}$

4. Two masses m_1 and m_2 are suspended together by a massless spring of constant K. When the masses are in equilibrium, m_1 is removed without disturbing the system. The amplitude of oscillations is:
(1) $\frac{m_1 g}{K}$ (2) $\frac{m_2 g}{K}$
(3) $\frac{(m_1+m_2)g}{K}$ (4) $\frac{(m_1-m_2)g}{K}$

5. In arrangement given in figure, if the block of mass m is displaced, the frequency is given by:



(1) $n=\frac{1}{2\pi}\sqrt{\left(\frac{k_1-k_2}{m}\right)}$
(2) $n=\frac{1}{2\pi}\sqrt{\left(\frac{k_1+k_2}{m}\right)}$
(3) $n=\frac{1}{2\pi}\sqrt{\left(\frac{m}{k_1+k_2}\right)}$
(4) $n=\frac{1}{2\pi}\sqrt{\left(\frac{m}{k_1-k_2}\right)}$

6. A weightless spring which has a force constant K oscillates with frequency n when a mass m is suspended from it. The spring is cut into two equal halves and a mass $2m$ is suspended from one half. The frequency of oscillation will now become:
(1) n
(2) $2n$
(3) $n/\sqrt{2}$
(4) $n(2)^{1/2}$

7. The time period of a simple pendulum of length L as measured in an elevator descending with acceleration $g/3$ is:
(1) $2\pi\sqrt{\frac{3L}{g}}$ (2) $\pi\sqrt{\left(\frac{3L}{g}\right)}$
(3) $2\pi\sqrt{\left(\frac{3L}{2g}\right)}$ (4) $2\pi\sqrt{\frac{2L}{3g}}$

8. A plate oscillated with time period 'T'. Suddenly, another plate put on the first plate, then time period:
(1) will decrease.
(2) will increase.
(3) will be same.
(4) None of these

9. The maximum speed of a particle executing S.H.M. is 1 m/s and its maximum acceleration is 1.57 m/sec^2. The time period of the particle will be:
(1) $\frac{1}{1.57}$ sec (2) 1.57 sec
(3) 2 sec (4) 4 sec

10. The kinetic energy of a particle, executing S.H.M. is 16 J, when it is in its mean position. If the amplitude of oscillations is 25 cm and the mass of the particle is 5.12 kg, the time period of its oscillation is:

(1) $\frac{\pi}{5}$ sec (2) 2π sec

(3) 20π sec (4) 5π sec

11. The displacement x (in *metres*) of a particle performing simple harmonic motion is related to time t (in *seconds*) as $x = 0.05\cos\left(4\pi t + \frac{\pi}{4}\right)$. The frequency of the motion will be:

(1) 0.5 Hz (2) 1.0 Hz

(3) 1.5 Hz (4) 2.0 Hz

12. The instantaneous displacement of a simple pendulum oscillator is given by $x = A\cos\left(\omega t + \frac{\pi}{4}\right)$. Its speed will be maximum at time.

(1) $\frac{\pi}{4\omega}$ (2) $\frac{\pi}{2\omega}$

(3) $\frac{\pi}{\omega}$ (4) $\frac{2\pi}{\omega}$

13. Which one of the following statements is **true**?

(1) Both light and sound waves in air are transverse.

(2) The sound waves in air are longitudinal while the light waves are transverse.

(3) Both light and sound waves in air are longitudinal.

(4) Both light and sound waves can travel in vacuum.

14. A point source emits sound equally in all directions in a non-absorbing medium. Two points *P* and *Q* are at distance of 2 m and 3 m respectively from the source. The ratio of the intensities of the waves at *P* and *Q* is:

(1) 9 : 4 (2) 2 : 3

(3) 3 : 2 (4) 4 : 9

15. Two vibrating tuning forks produce progressive waves given by $y_1 = 4\sin 500\pi t$ and $y_2 = 2\sin 506\pi t$. Number of beats produced per minute is:

(1) 360

(2) 180

(3) 3

(4) 60

16. The driver of a car travelling with speed 30 m/sec towards a hill sounds a horn of frequency 600 Hz. If the velocity of sound in air is 330 m/s, the frequency of reflected sound as heard by driver is:

(1) 555.5 Hz (2) 720 Hz

(3) 500 Hz (4) 550 Hz

17. A string is stretched between fixed points separated by 75.0 cm. It is observed to have resonant frequencies of 420 Hz and 315 Hz. There are no other resonant frequencies between these two. The lowest resonant frequency for this string is:

(1) 105 Hz (2) 155 Hz

(3) 205 Hz (4) 10.5 Hz

18. If n_1, n_2 and n_3 are the fundamental frequencies of three segments into which a string is divided, then the original fundamental frequency n of the string is given by:

(1) $\frac{1}{n} = \frac{1}{n_1} + \frac{1}{n_2} + \frac{1}{n_3}$

(2) $\frac{1}{\sqrt{n}} = \frac{1}{\sqrt{n_1}} + \frac{1}{\sqrt{n_2}} + \frac{1}{\sqrt{n_3}}$

(3) $\sqrt{n} = \sqrt{n_1} + \sqrt{n_2} + \sqrt{n_3}$

(4) $n = n_1 + n_2 + n_3$

19. If we study the vibration of a pipe open at both ends, then the following statement is **not** true:

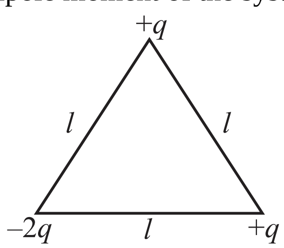
(1) odd harmonics of the fundamental frequency will be generated.

(2) all harmonics of the fundamental frequency will be generated.

(3) pressure change will be maximum at both ends.

(4) open end will be antinode.

20. Electric dipole moment of the system is:



(1) $\sqrt{3}ql$

(2) ql

(3) $\sqrt{2}ql$

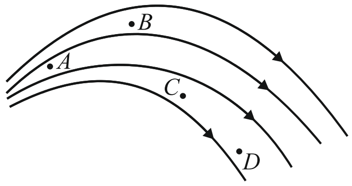
(4) $2ql$

21. The electric field is given by $\vec{E} = (5\hat{i} + 8\hat{j} + 9\hat{k})$, the electric flux through surface, whose area vector is given by $\vec{A} = 10\hat{i}$, would be:

(1) 5 units (2) 10 units

(3) 50 units (4) 100 units

22. A positive test charge is released at different points in the following field. At which point the acceleration of the test charge is minimum?



(1) A

(2) B

(3) C

(4) D

23. Twenty seven drops of same size are charged at 220 V each They combine to form a bigger drop. Calculate the potential of the bigger drop.

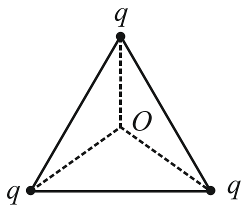
(1) 1320 V

(2) 1520 V

(3) 1980 V

(4) 660 V

24. Three isolated equal charges are placed at the three corners of an equilateral triangle as shown in figure. The statement which is true for net electric potential V and net electric field intensity E at the centre of the triangle is:



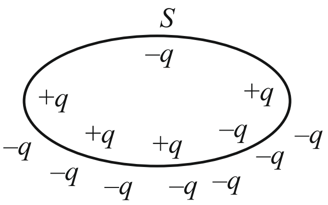
(1) $E = 0, V = 0$

(2) $V = 0, E \neq 0$

(3) $V \neq 0, E = 0$

(4) $V \neq 0, E \neq 0$

25. The flux of electric field due to these charges through the surface S is:



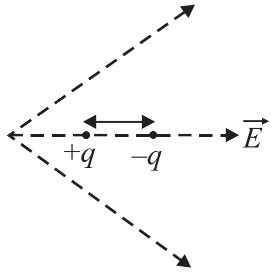
(1) $\dfrac{2q}{\varepsilon_0}$

(2) $\dfrac{3q}{\varepsilon_0}$

(3) $\dfrac{-5q}{\varepsilon_0}$

(4) $q\varepsilon_0$

26. A dipole is placed in an electric field as shown, in which direction will it move?



(1) Towards the left as its potential energy will increase.

(2) Towards the right as its potential energy will decrease.

(3) Towards the left as its potential energy will decreases.

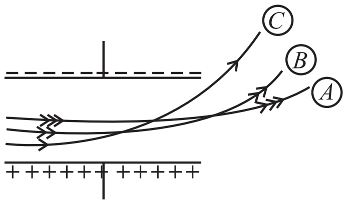
(4) Towards the right as its potential energy will increases.

27. If two charges each of charge 2 μC are located at 3 m apart, then the electrostatic force acting between them is:

(1) 2×10^{-2} N (2) 2×10^{-3} N

(3) 4×10^{-3} N (4) 6×10^{-3} N

28. Three particles are projected in a uniform electric field with same velocity perpendicular to the field as shown. Which particle has highest charge to mass ratio?



(1) A

(2) B

(3) C

(4) All have same charge to mass ratio.

29. The parallel infinite line charges with linear charge densities +λ C/m –λ C/m are placed at a distance of $2R$ in free space. What is the electric field mid-way between the two line charges.

(1) Zero (2) $\dfrac{2\lambda}{\pi\varepsilon_0 R}$ N/C

(3) $\dfrac{\lambda}{\pi\varepsilon_0 R}$ N/C (4) $\dfrac{\lambda}{2\pi\varepsilon_0 R}$ N/C

30. The electric potential V at a point $P(x, y, z)$ in space is given by $V = 4x^2$ volt. Electric field at a point (1 m, 0, 2 m) in V/m is:

(1) 8 along –ve x-axis

(2) 8 along +ve x-axis

(3) 16 along –ve x-axis

(4) 16 along +ve x-axis

31. Given below are two statements:

Statement I: When some charge is given to an irregular shaped conductor, it distributes itself so that charge density is same everywhere.

Statement II: A conductor has to be equipotential under static condition.

In the light of the above statements, choose the most appropriate answer from the options given below:

(1) Statement I is correct but Statement II is incorrect.
(2) Statement I is incorrect but Statement II is correct.
(3) Both Statement I and Statement II are correct.
(4) Both Statement I and Statement II are incorrect.

32. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:

Assertion A: Electric lines of force cross each other.

Reason R: The resultant electric field at a point is the superimposition of the electric fields at that point.

In the light of the above statements, choose the **correct** answer from the options given below:

(1) A is true but R is false.
(2) A is false but R is true.
(3) Both A and R are true and R is the correct explanation of A.
(4) Both A and R are true but R is not the correct explanation of A.

33. Given below are two statements: one is labelled as

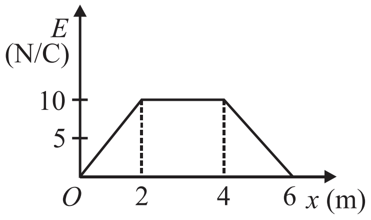
Assertion A: In a non-uniform electric field, a dipole will have translatory as well as rotatory motion.

Reason R: In a non-uniform electric field, a dipole experiences a force as well as torque.

In the light of the above statements, choose the **correct** answer from the options given below:

(1) A is true but R is false.
(2) A is false but R is true.
(3) Both A and R are true and R is the correct explanation of A.
(4) Both A and R are true but R is NOT the correct explanation of A.

34. Figure shows the variation of electric field intensity E versus distance x. What is the magnitude of potential difference between the points at $x = 2$ m and at $x = 6$ m from O?



(1) 30 V (2) 60 V
(3) 40 V (4) 80 V

35. Match **List-I** with **List-II:**

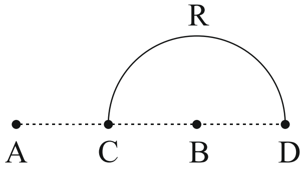
List-I		List-II	
(A)	Coulomb's law	(I)	Total electric flux through a closed surface.
(B)	Gauss's law	(II)	Vector sum of forces.
(C)	Principle of superposition	(III)	Force is inversely proportional to square of distance
(D)	Quantisation of charge	(IV)	Discrete nature of charge

Choose the **correct** answer from the options given below:

(1) A-II, B-III, C-I, D-IV
(2) A-III, B-I, C-II, D-IV
(3) A-I, B-IV, C-III, D-II
(4) A-I, B-II, C-II, D-IV

SECTION - B

36. Charges + q and – q are placed at points A and B respectively which are at a distance $2L$ apart, C is the mid-point between A and B, The work done in moving a charge $+Q$ along the semicircle CRD is:



(1) $\frac{qQ}{4\pi\varepsilon_0 L}$ (2) $-\frac{qQ}{4\pi\varepsilon_0 L}$

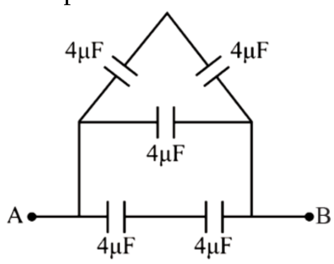
(3) $\frac{qQ}{6\pi\varepsilon_0 L}$ (4) $-\frac{qQ}{6\pi\varepsilon_0 L}$

37. A parallel plate capacitor has the space between its plates filled by the two slabs of thickness $d/2$ each and dielectric constants K_1 and K_2, d is the plate separation of capacitors. The capacity of the capacitor is:

(1) $\frac{2\varepsilon_0 d}{A}\left(\frac{K_1+K_2}{K_1K_2}\right)$ (2) $\frac{2\varepsilon_0 A}{d}\left(\frac{K_1K_2}{K_1+K_2}\right)$

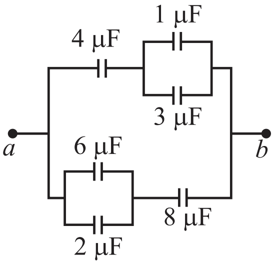
(3) $\frac{2\varepsilon_0 A}{d}(K_1+K_2)$ (4) $\frac{2\varepsilon_0 A}{d}\left(\frac{K_1+K_2}{K_1K_2}\right)$

38. Equivalent capacitance between A and B is:



(1) 8 μF (2) 6 μF
(3) 26 μF (4) 10/3 μF

39. The equivalent capacitance between *a* and *b* for the combination of capacitors shown in the figure is:



(1) 6.0 μF (2) 4.0 μF
(3) 2.0 μF (4) 3.0 μF

40. A parallel plate capacitor of plate area A and plate separation d is filled with a dielectric of dielectric constant $k = 2$ and thickness $d/3$. The effective capacitance of the capacitor will be:

(1) $6\frac{A\varepsilon_0}{d}$ (2) $3\frac{A\varepsilon_0}{2d}$

(3) $\frac{6}{5}\frac{A\varepsilon_0}{d}$ (4) $\frac{5}{3}\frac{A\varepsilon_0}{d}$

41. A source of unknown frequency gives 4 beats/s, when sounded with a source of known frequency 250 Hz. The second harmonic of the source of unknown frequency gives five beats per second, when sounded with a source of frequency 513 Hz. The unknown frequency is:
(1) 246 Hz (2) 240 Hz
(3) 260 Hz (4) 254 Hz

42. A wave travelling in the positive x-direction having displacement along y-direction as 1 m, wavelength $2\pi m$ and frequency of $\frac{1}{\pi}$ Hz is represented by:
(1) $y = \sin(2\pi x - 2\pi t)$
(2) $y = \sin(10\pi x - 20\pi t)$
(3) $y = \sin(2\pi x + 2\pi t)$
(4) $y = \sin(x - 2t)$

43. The equation of a simple harmonic wave is given by $y = 3\sin\frac{\pi}{2}(50t - x)$, where x and y are in metres and t is in seconds. The ratio of maximum particle velocity of the wave velocity is:

(1) 2π (2) $\frac{3}{2}\pi$

(3) 3π (4) $2/3\ \pi$

44. A tuning fork with frequency 800 Hz produces resonance in a resonance column tube with upper end open and lower end closed by water surface. Successive resonance are observed at lengths 9.75 cm, 31.25 cm and 52.75 cm. The speed of sound in air is:
(1) 172 m/s (2) 500 m/s
(3) 156 m/s (4) 344 m/s

45. A tuning fork is used to produce resonance in a glass tube. The length of the air column in this tube can be adjusted by a variable piston. At room temperature of 27ºC two successive resonances are produced at 20 cm and 73 cm of column length. If the frequency of the tuning fork is 320 Hz, the velocity of sound in air at 27ºC is:
(1) 330 m/s (2) 339 m/s
(3) 300 m/s (4) 350 m/s

46. The two nearest harmonics of a tube closed at one end and open at other end are 220 Hz and 260 Hz. What is the fundamental frequency of the system?
(1) 10 Hz
(2) 20 Hz
(3) 30 Hz
(4) 40 Hz

47. Three sound waves of equal amplitudes have frequencies $(n - 1)$, n, $(n + 1)$. They superimpose to give beats. The number of beats produced per second will be:
(1) 1
(2) 4
(3) 3
(4) 2

48. A uniform rope of length L and mass m_1 hangs vertically from a rigid support. A block of mass m_2 is attached to the free end of the rope. A transverse pulse of wavelength λ_1 is produced at the lower and of the rope. The wavelength of the pulse when it reaches the top of the rope is λ_2. The ratio λ_2/λ_1 is:

(1) $\sqrt{\frac{m_1 + m_2}{m_1}}$ (2) $\sqrt{\frac{m_1}{m_2}}$

(3) $\sqrt{\frac{m_1 + m_2}{m_2}}$ (4) $\sqrt{\frac{m_2}{m_1}}$

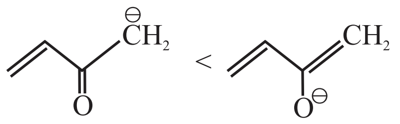
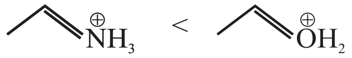
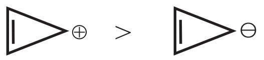
49. The fundamental frequency of a closed organ pipe of length 20 cm is equal to the second overtone of an organ pipe open at both the ends. The length of organ pipe open at both the ends is:
(1) 140 cm
(2) 80 cm
(3) 100 cm
(4) 120 cm

50. Equation of a transverse wave pulse is $y = \frac{9}{3 + (x - 10t)^2}$, (where x and y are in metre and t is in second). The speed of the wave pulse is:
(1) 3 m/s
(2) 5 m/s
(3) 10 m/s
(4) 100 m/s

SECTION-(II)CHEMISTRY

SECTION - A

51. The correct IUPAC name from the **incorrect** name 4-Amino-3-hydroxy-2-butene is:
(1) 1-Amino-2-hydroxybut-2-ene
(2) 4-Aminobut-2-en-3-ol
(3) 1-Aminobut-2-en-2-ol
(4) 1-Aminobut-2-enol

52. Choose the **correct** comparison of stability:
(1) 
(2) 
(3) 
(4) Both (1) and (3)

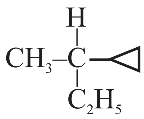
53. n-pentane and neopentane are:
(1) Functional isomers
(2) Geometrical isomers
(3) Chain isomers
(4) Position isomers

54. Which of the following is the strongest –I group:
(1) $-\overset{+}{N}(CH_3)_3$ (2) $-\overset{+}{N}H_3$
(3) $-\overset{+}{S}(CH_3)_2$ (4) $-F$

55. The IUPAC name for,
$$CH_3CHOHCH_2-\underset{CH_3}{\overset{CH_3}{\underset{|}{\overset{|}{C}}}}-OH$$ is
(1) 1,1-Dimethylbutane-1,2-diol
(2) 2-Methylpentane-2,4-diol
(3) 4-Methylpentane-2,4-diol
(4) 1,3,3-Trimethypropane-1,3-diol.

56. Read the three statements:
Statement I: H.C (Hyperconjugation) is no bond resonance.
Statement II: In $CH_2 = CH_2$, Hyperconjugation is not possible
Statement III: Hyper conjugation is also called as (σ) - (σ) conjugation
(1) All the three statement are correct
(2) Only statement (I) is correct
(3) Statement (I) and statement (II) are correct
(4) Only statement (II) is correct

57. Which hybridised state of carbon exerts strongest –I effect having how much percentage of 's' character?
(1) 25% (2) 33%
(3) 50% (4) Equal in all

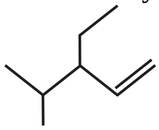
58. Amongst the following compounds, the optically active alkane is:
(1) $CH_3–CH_2–CH_3$
(2) 
(3) $CH_3-CH_2-\overset{CH_3}{\overset{|}{C}H}-CH_3$
(4) $CH_3–CH_2–C≡C–H$

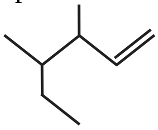
59. The state of hybridization of C_2,C_3,C_5 and C_6 of the hydrocarbon,
$$\underset{7}{CH_3}-\underset{CH_3}{\overset{CH_3}{\underset{6|}{\overset{|}{C}}}}-\underset{5}{C}H=\underset{4}{C}H-\overset{CH_3}{\overset{|}{\underset{3}{C}H}}-\underset{2}{C}\equiv\underset{1}{C}H$$
is in the following sequence.
(1) sp^3, sp^2, sp^2 and sp
(2) sp, sp^2, sp^2 and sp^3
(3) sp, sp^2, sp^3 and sp^2
(4) sp, sp^3, sp^2 and sp^3

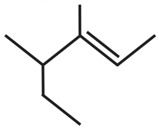
60. Given below are two statements:
Statement I: Nucleophiles attack the regions of high electron density.
Statement II: Nucleophiles act as Lewis base.
In the light of the above statements, choose the most appropriate answer from the options given below:
(1) Statement I is correct but Statement II is incorrect.
(2) Statement I is incorrect but Statement II is correct.
(3) Both Statement I and Statement II are correct.
(4) Both Statement I and Statement II are incorrect.

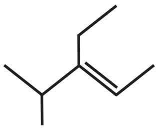
61. Which of the following pair is **not** a functional isomer of each other?
(1) Acetic acid and Methyl formate
(2) Propionic acid and Methylacetate
(3) Propanal and Propan-2-one
(4) 1-Chloropropane and 2-Chloropropane

62. Which of the following is the **correct** bond line formula of 3-Ethyl-4-methylpent-1-ene?

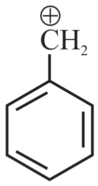
(1) 

(2) 

(3) 

(4) 

63. The order of stability of the following carbocations is

$CH_3=CH-\overset{\oplus}{C}H_2$; $CH_3-CH_2-\overset{\oplus}{C}H_2$; 

I II III

(1) III > I > II (2) III > II > I
(3) II > III > I (4) I > II > III

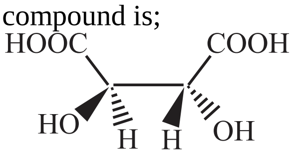
64.

CH_2-CO_2H
$CH-CO_2H$
$HO-CH-CO_2H$

In the given compound number of chiral centre and stereoisomer are:

(1) 1, 2 (2) 2, 4
(3) 2, 3 (4) 3, 8

65. The absolute configuration of the following compound is;



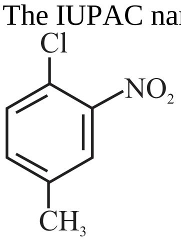
(1) R, R (2) R, S
(3) S, R (4) S, S

66. Select the incorrect statement from the following option.

(A) Meso compound possess both plane of symmetry and centre of symmetry.
(B) Meso compound possess either plane or centre of symmetry.
(C) Meso compound does not possess either plane or centre of symmetry.
(D) Meso compounds are externally compensated form.

(1) Only A (2) Only B
(3) A, C and D only (4) A, B and C only

67. The IUPAC name of the following compound is:

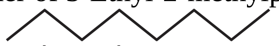


(1) 1-Chloro-2-nitro-4-methylbenzene
(2) 1-Chloro-4-methyl-2-nitrobenzene
(3) 2-Chloro-1-nitro-5-methylbenzene
(4) m-Nitro-p-chlorotoluene

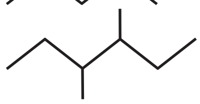
68. The number of sigma (σ) and pi (π) bonds in pent-2-en-4-yne is;

(1) 10 σ bonds and 3π bonds
(2) 8 σ bonds and 5π bonds
(3) 11 σ bonds and 2π bonds
(4) 13 σ bonds and noπ bonds

69. Which of the following substances is **not** an isomer of 3-Ethyl-2-methylpentane?

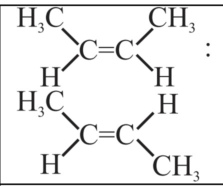
(1) 

(2) 

(3) 

(4) All are isomers

70. Match **List-I** with **List-II**:

	List-I		**List-II**
(A)	A pair of functional isomers	(I)	
(B)	A pair of geometrical isomers	(II)	C_2H_5CHO : CH_3COCH_3
(C)	A pair of metamers	(III)	$CH_3OC_3H_7$: $C_2H_5OC_2H_5$
(D)	A pair of tautomers	(IV)	$H_2C = CHOH$: CH_3CHO

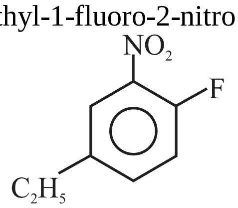
Choose the **correct** answer from the options given below:

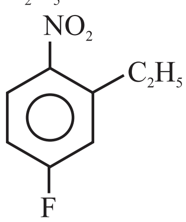
(1) A-I, B-II, C-III, D-IV
(2) A-II, B-I, C-III, D-IV
(3) A-III, B-I, C-II, D-IV
(4) A-IV, B-III, C-I, D-II

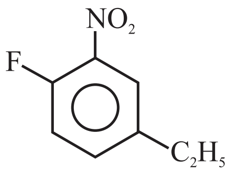
71. The number of 1°, 2° and 3° carbon atoms present in isopentane are respectively:

(1) 3, 2, 1 (2) 2, 3, 1
(3) 3, 1, 1 (4) 2, 2, 1

72. The **correct** structural formula of the compound 4-Ethyl-1-fluoro-2-nitro benzene is:

(1) 

(2) 

(3) 

(4) Both (1) and (3)

73. Structure of 4-Methylpent-2-enal is:

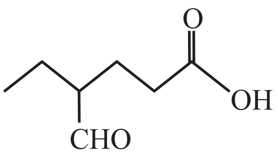
(1) $CH_2=\underset{H}{C}-\overset{CH_3}{\underset{H}{C}}-CH_2-\overset{O}{\overset{\|}{C}}-H$

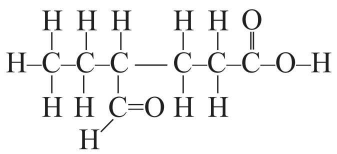
(2) $CH_3-CH_2-\underset{CH_3}{C}=CH-\overset{O}{\overset{\|}{C}}-H$

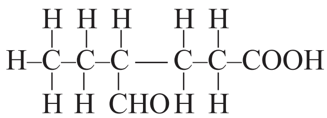
(3) $CH_3-CH_2-CH=\underset{CH_3}{C}-\overset{O}{\overset{\|}{C}}-H$

(4) $CH_3-\underset{CH_3}{CH}-CH=CH-\overset{O}{\overset{\|}{C}}-H$

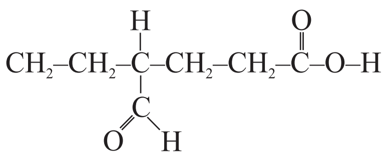
74. The complete structural formula of the given bond-line formula is:



(1) 

(2) 

(3) $CH_3-CH_2-\underset{CHO}{CH}-CH_2-CH_2-COOH$

(4) 

75. Given below are two statement: one is labelled as Assertion A and the other is labelled as Reason R:

Assertion (A): Tertiary carbocations are generally formed more easily than primary carbocations.

Reason (R): Hyperconjugation as well as inductive effect due to additional alkyl groups stabilize tertiary carbocations.

In the light of the above statements, choose the **correct** answer from the options given below:

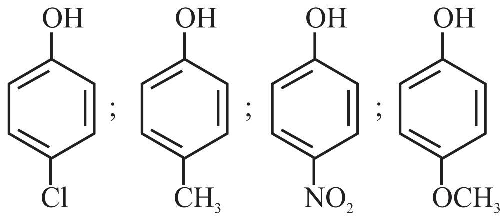
(1) Both A and R are true and R is the correct explanation of A.

(2) Both A and R are true but R is NOT the correct explanation of A.

(3) A is true but R is false.

(4) A is false but R is true.

76. Arrange the following compounds in order of decreasing acidity:



(I) (II) (III) (IV)

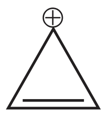
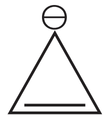
(1) II > IV > I > III

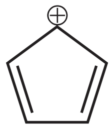
(2) I > II > III > IV

(3) III > I > II > IV

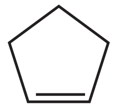
(4) IV > III > I > II

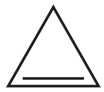
77. Which of the following is most stable?

(1)  (2) 

(3)  (4) 

78. Which of the following hydrocarbon is most acidic?

(1)  (2) 

(3)  (4) 

79. An isomer of C_6H_{14} forms three monochloro derivatives. The isomer may be: (Excluding stereo isomer)

(1) neo-Pentane

(2) n-Hexane

(3) 2, 3-Dimethylbutane

(4) iso-Hexane

80. Oxidation of isobutylene with acidic potassium permanganate gives:

(1) Acetone + CO_2

(2) Acetic acid

(3) Acetic acid + CO_2

(4) Acetic acid + acetone

81. During dehydration of alcohols to alkenes by heating with conc. H_2SO_4 the initiation step is:

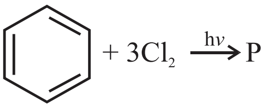
(1) Protonation of alcohol molecule

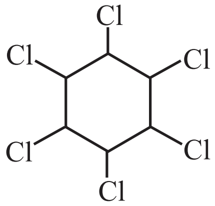
(2) Formation of carbocation

(3) Elimination of water

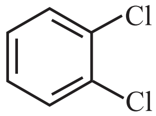
(4) Formation of an ester

82. What will be the final product (P) of the following reaction?



(1) 

(2) 

(3) 

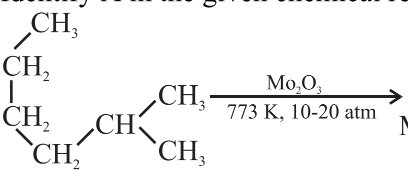
(4) None of these

83. When 1, 1, 2, 2-tetrabromopropane is heated with zinc powder in alcohol, which is formed:

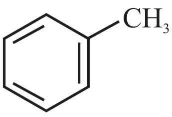
(1) Propyne (2) Propene

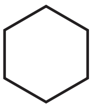
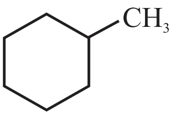
(3) Propane (4) Propadiene

84. Identify A in the given chemical reaction.

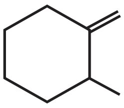
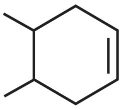


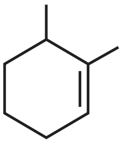
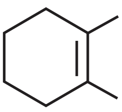
(A) Major product

(1)  (2) 

(3)  (4) 

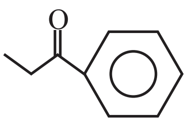
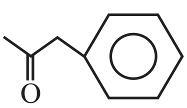
85. In which of the given compound after the catalytic hydrogenation with H_2 less heat will be evolved:

(1)  (2) 

(3)  (4) 

SECTION-B

86. $C_6H_5-C\equiv C-CH_3 \xrightarrow[H_2SO_4]{HgSO_4}$ Product is:

(1)  (2) 

(3) $C_6H_5-\underset{OH}{\underset{|}{C}}=CHCH_3$ (4) $C_6H_5-CH=\underset{OH}{\underset{|}{C}}-CH_3$

87. Given below are two statements:

Statement I: The melting point of neo-pentane is greater than that of n-pentane but the boiling point of n-pentane is more than that of neo- pentane.

Statement II: Melting point depends upon packing in crystal lattice whereas boiling point depends upon the surface area of the molecule.

In the light of the above statements, choose the most appropriate answer from the options given below:

(1) Statement I is correct but Statement II is incorrect.

(2) Statement I is incorrect but Statement II is correct.

(3) Both Statement I and Statement II are correct.

(4) Both Statement I and Statement II are incorrect.

88. Soda lime decarboxylation of sodium butanoate produces;

(1) Propane (2) Ethane

(3) Butane (4) Methane

89. 2-Hexene $\xrightarrow[(ii)H_2O]{(i)O_3}$ Products

The two products formed in above reaction are:

(1) Butanoic acid and acetic acid

(2) Butanal and acetic acid

(3) Butanal and acetaldehyde

(4) Butanoic acid and acetaldehyde

90. The reaction

$CH_4 + Cl_2 \xrightarrow{\text{UV light}} CH_3Cl + HCl$

is an example of:

(1) Addition reaction

(2) Substitution reaction

(3) Elimination reaction

(4) Rearrangement reaction

91. Cyclohexene on reaction with cold dilute $KMnO_4$ forms;

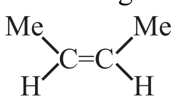
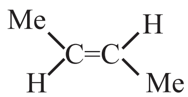
(1) trans-Cyclohexane-1,2-diol

(2) Hexanoic acid

(3) cis-Cyclohexane-1,2-diol

(4) None of these

92. Out of the given two isomers which property for second is greater than first;

(I) (II)

(1) Dipole moment (2) Boiling point

(3) Solubility in H_2O (4) Melting point

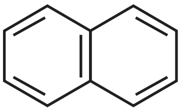
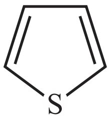
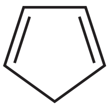
93. Match **List-I** with **List-II**:

List-I		**List-II**	
(A)	Decarboxylation	(I)	Paraffins
(B)	Wurtz reaction	(II)	Alkyl halide + $\xrightarrow{\text{Na/ether}}$
(C)	$CH_4 + Cl_2 \xrightarrow{\text{sunlight}} CH_3Cl$	(III)	RCOONa + Sodalime
(D)	Saturated hydrocarbons	(IV)	Free radical substitution

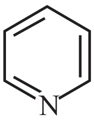
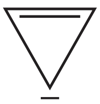
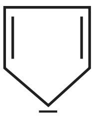
Choose the **correct** answer from the options given below:

(1) A-III, B-II, C-I, D-IV
(2) A-I, B-II, C-III, D-IV
(3) A-III, B-II, C-IV, D-I
(4) A-IV, B-III, C-II, D-I

94. Among the following which is non aromatic in nature.

(1) 
(2) 
(3) 
(4) 

95. Which of the following compounds is **not** aromatic according to the Huckle's rule?

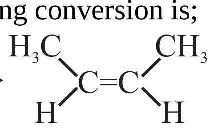
(1) 
(2) 
(3) 
(4) 

96. $CH_3{-}\underset{\displaystyle OH}{\underset{|}{C}H}{-}CH_2{-}CH_3 \xrightarrow[H_2SO_4]{\text{Conc.}} \text{Product (I)} + \text{Product (II)}$

Which is **not** true regarding the products?

(1) Product-I and II are position isomers.
(2) Product-I and II contains the same number of sp^3 and sp^2 carbon atoms.
(3) The yield of the product I and II is same.
(4) Reaction obeys Saytzeff rule.

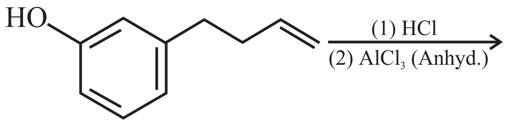
97. For the conversion of A into B the most suitable reagent for the following conversion is;

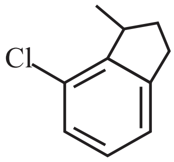
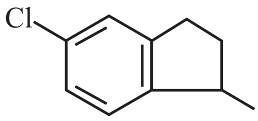
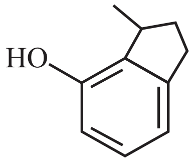
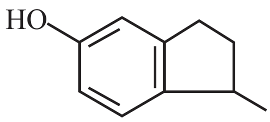
$CH_3{-}C{\equiv}C{-}CH_3 \longrightarrow$ 

(A) cis-But-2-ene (B)

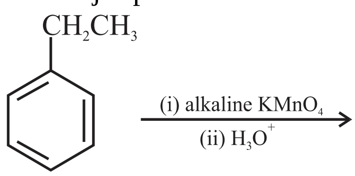
(1) Na/liquid NH_3
(2) H_2, Pd/C, quinoline
(3) Zn/HCl
(4) Hg^{2+}/H^+, H_2O

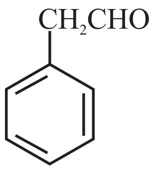
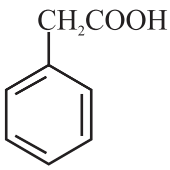
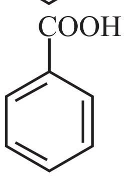
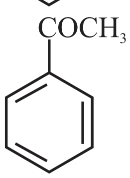
98. The major product of the following reaction is:



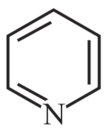
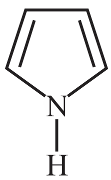
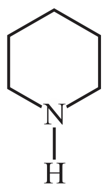
(1) 
(2) 
(3) 
(4) 

99. The major product of the following reaction is:



(1) 
(2) 
(3) 
(4) 

100. Arrange the following amines in the decreasing order of basicity.

(I) (II) (III)

(1) I > III > II
(2) III > I > II
(3) III > II > I
(4) I > II > III

SECTION-(III) BOTANY

SECTION - A

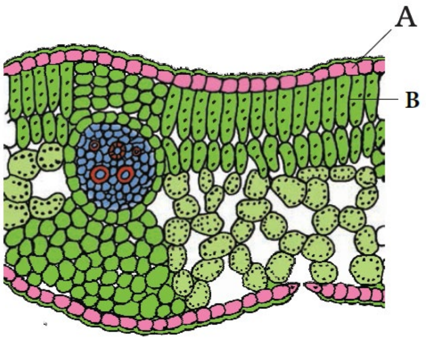
101. All the xylem elements, when mature, are dead **except:**
(1) tracheids.
(2) vessels.
(3) xylem parenchyma.
(4) xylem fibres.

102. Select the **mismatched** pair.
(1) Guard cells – Opening and closing of stomata
(2) Bulliform cells – Empty colourless cell
(3) Stomata – Unicellular cells for water absorption
(4) Epidermis–Outermost layer of root, stem and leaf

103. Pericycle in dicotyledonous stem is in the form of:
(1) circular patches of sclerenchyma.
(2) semi-lunar patches of sclerenchyma.
(3) semi-lunar patches of collenchyma.
(4) circular patches of collenchyma.

104. Given below are two statements:
Statement I: Different organs in a plant show differences in their internal structure.
Statement II: Internal structures in plants show adaptations to diverse environments.
In the light of the above statements, choose the most appropriate answer from the options given below:
(1) Statement I is correct but Statement II is incorrect.
(2) Statement I is incorrect but Statement II is correct.
(3) Both Statement I and Statement II are correct.
(4) Both Statement I and Statement II are incorrect.

105. Transverse section of dicotyledonous leaf passing through the midrib is given below. Identify **A** and **B** and select the **correct** labeling for **A** and **B**.



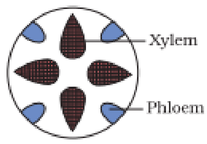
(1) A–Epidermis, B–Spongy mesophyll
(2) A– Epidermis, B– Palisade mesophyll
(3) A– Epidermis, B – Guard cells
(4) A– Cuticle, B – Phloem

106. Select the **correct** option from the following regarding dicotyledonous root.
(1) Lateral roots develop from endodermis.
(2) Endodermis is the outermost layer of cortex.
(3) Endodermis possess intercellular spaces.
(4) Endodermis is single layered.

107. Collenchyma differs from sclerenchyma in:
(1) retaining protoplasm at maturity.
(2) having thick walls.
(3) having a wide lumen.
(4) being meristematic.

108. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:
Assertion A: Ground tissue consists of complex tissues such as parenchyma, collenchyma and sclerenchyma.
Reason R: Ground tissue includes pericycle and pith.
In the light of the above statements, choose the **correct** answer from the options given below:
(1) A is true but R is false.
(2) A is false but R is true.
(3) Both A and R are true and R is the correct explanation of A.
(4) Both A and R are true but R is NOT the correct explanation of A.

109. Identify from the following where do we find this type of vascular bundle.



(1) Dicotyledonous root
(2) Monocotyledonous stem
(3) Monocotyledonous leaf
(4) Dicotyledonous leaf

110. Select the **correct** statement regarding epidermis.
A. Epidermal tissue system forms the outer-most covering of the whole plant body.
B. It is made up of elongated cells.
C. It is always multi-layered.
D. Epidermal cells have a large vacuole.
E. Epidermal cells are parenchymatous.
Choose the most appropriate answer from the options given below:
(1) B, C and D only
(2) A, B, C and D
(3) A, B, D and E only
(4) B and D only

111. The parenchymatous cells in dicotyledonous roots which lie between the xylem and the phloem are called:
(1) conjuctive tissue.
(2) cambium.
(3) casparian strip.
(4) endodermis.

112. Match **List-I** with **List-II**.

List-I		List-II	
(A)	Cuticle	(I)	Pore
(B)	Trichomes	(II)	Unicellular
(C)	Root hair	(III)	Waxy layer
(D)	Stomata	(IV)	Multicellular

Choose the **correct** answer from the options given below:
(1) A-II, B-III, C-I, D-IV
(2) A-IV, B-III, C-II, D-I
(3) A-III, B-IV, C-II, D-I
(4) A-II, B-I, C-III, D-IV

113. Casparian strips are the bands of thickenings present in:
(1) radial wall of the dicotyledonous root.
(2) central wall of the dicotyledonous stem.
(3) tangential wall of the dicotyledonous leaf.
(4) Both (1) and (3)

114. Identify from the following which is/are related to dicotyledonous leaf.
A. Parallel venation.
B. Absence of bulliform cells from epidermis.
C. Mesophyll not differentiated into palisade and spongy.
D. Well differentiated mesophyll.
Choose the most appropriate answer from the options given below:
(1) A, B and D are correct
(2) A, B and C are incorrect
(3) A, B and C are correct
(4) B and D are correct

115. Ground tissue in monocotyledonous stem is made up of:
(1) parenchyma cells.
(2) collenchyma cells.
(3) sclerenchyma cells.
(4) tracheids.

116. Vascular bundles in leaves are surrounded by:
(1) bundle sheath cells.
(2) endodermis.
(3) epidermis.
(4) hypodermis.

117. Pith in dicotyledonous stem is composed of:
(1) parenchymatous cells.
(2) sclerenchymatous cells.
(3) collenchymatous cells.
(4) Both (2) and (3)

118. Given below are two statements:
Statement I: Glucose and fructose are phosphorylated to give rise to glucose-6-phosphate.
Statement II: The phosphorylated form of glucose isomerises to produce fructose-6-phosphate.
In the light of the above statements, choose the most appropriate answer from the options given below:
(1) Statement I is correct but Statement II is incorrect.
(2) Statement I is incorrect but Statement II is correct.
(3) Both Statement I and Statement II are correct.
(4) Both Statement I and Statement II are incorrect.

119. Read the following statements and select the **incorrect** option regarding respiration.
(1) The breaking of the C-C bonds of complex compounds occurs in respiration.
(2) The compounds that are reduced during this process are known as respiratory substrates.
(3) Energy is released during respiration.
(4) Energy contained in respiratory substrates is not released in a single step.

120. How many total ATP molecules are produced through substrate level phosphorylation in TCA cycle, if a molecule of glucose is used as respiratory substrate?
(1) 2 (2) 8
(3) 12 (4) 5

121. Identify from the following the **correct** statement related to oxidative decarboxylation**.**
A. ATP is generated.
B. NADH + H^+ is formed.
C. Utilisation of acetyl CoA occurs.
D. Mg^{+2} ion is used.
Choose the most appropriate answer from the options given below:
(1) B, C and D only
(2) B and D only
(3) A only
(4) A, B, C and D

122. Which of the given acts as a mobile carrier for transfer of electrons between complex III and IV in Electron Transport System (ETS) of mitochondria?

(1) Cytochrome *c*
(2) Ubiquinone
(3) Cytochrome a_1
(4) Cytochrome a_3

123. Match **List-I** with **List-II**.

List-I		List-II	
(A)	Complex I	(I)	Cytochrome bc_1 complex
(B)	Complex II	(II)	NADH dehydrogenase
(C)	Complex III	(III)	Succinate dehydrogenase
(D)	Complex IV	(IV)	Cytochrome *c* oxidase

Choose the **correct** answer from the options given below:

(1) A-I, B-III, C-II, D-IV
(2) A-IV, B-III, C-II, D-I
(3) A-II, B-III, C-I, D-IV
(4) A-II, B-I, C-III, D-IV

124. Select the **incorrect** statement from the following.

(1) NADH is oxidised to NAD^+ slowly in fermentation.
(2) The reaction of oxidation of NADH to NAD^+ is very vigorous in case of aerobic respiration.
(3) Fermentation accounts for partial breakdown of glucose.
(4) In anaerobic respiration CO_2 and H_2O are formed.

125. The connecting link between EMP pathway and citric acid cycle is:

(1) citric acid.
(2) glucose.
(3) succinic acid.
(4) acetyl CoA.

126. Select the **incorrect** statement w.r.t acetyl CoA formation from pyruvate.

(1) Reaction is catalysed by pyruvate dehydrogenase.
(2) $FADH_2$ is utilised in this process.
(3) This process cannot occur in muscle cell under inadequate oxygen supply.
(4) Two acetyl CoA are formed from one molecule of glucose.

127. During aerobic respiration, the complete oxidation of pyruvate by the stepwise removal of all the hydrogen atoms, leave:

(1) two molecules of CO_2.
(2) three molecules of O_2.
(3) four molecules of CO_2.
(4) three molecules of CO_2.

128. Match **List-I** with **List-II**.

List-I		List-II	
(A)	Acetyl CoA	(I)	4-Carbon compound
(B)	Fructose	(II)	2-Carbon compound
(C)	Pyruvic acid	(III)	6-Carbon compound
(D)	Oxaloacetic acid	(IV)	3-Carbon compound

Choose the **correct** answer from the options given below:

(1) A-III, B-II, C-I, D-IV
(2) A-I, B-III, C-II, D-IV
(3) A-II, B-III, C-I, D-IV
(4) A-II, B-III, C-IV, D-I

129. Identify from the following where do we find the enzymes of TCA cycle.

A. outer membrane of mitochondria.
B. mitochondrial intermembrane space.
C. mitochondrial matrix.
D. inner membrane of mitochondria.

Choose the most appropriate answer from the options given below:

(1) B, C and D only
(2) B and C only
(3) C and D only
(4) A, B, C and D

130. Which of the following is **not correct** for hexokinase?

(1) It helps in breakdown of glucose to fructose.
(2) It is essential for glycolysis.
(3) ATP is required for its activity.
(4) It is present in both aerobic and anaerobic organism.

131. Which of the following are involved in the first step of the Krebs' cycle?

(1) Oxaloacetic acid, pyruvic acid, water
(2) Acetyl CoA, water, isocitric acid
(3) Malic acid, oxaloacetic acid, citric acid
(4) Acetyl CoA, oxaloacetic acid, water

132. Number of steps required in glycolysis are:
(1) one.
(2) five.
(3) ten.
(4) fifteen.

133. Match **List-I** with **List-II**.

List-I		List-II	
(A)	Succinate dehydrogenase	(I)	Contains two coppers centres
(B)	Cytochrome *c* oxidase complex	(II)	Attached to the outer surface of inner mitochondrial membrane
(C)	Cytochrome *c*	(III)	Provides reducing equivalents to ubiquinone
(D)	Complex V	(IV)	ATP synthase

Choose the **correct** answer from the options given below:
(1) A-II, B-III, C-I, D-IV
(2) A-IV, B-III, C-II, D-I
(3) A-II, B-III, C-IV, D-I
(4) A-III, B-I, C-II, D-IV

134. Select the **correct** match from the following.
(1) Trichome - Fundamental tissue system
(2) Guard cells - Conducting tissue system
(3) Xylem - Epidermal tissue system
(4) Root hair - Epidermal tissue system

135. Given below are two statements:
Statement I: Yeasts poison themselves to death when the concentration of alcohol reaches about 13 per cent.
Statement II: Pyruvic acid dehydrogenase and alcohol dehydrogenase catalyse the reactions of fermentation in yeast.
In the light of the above statements, choose the most appropriate answer from the options given below:
(1) Statement I is correct but Statement II is incorrect.
(2) Statement I is incorrect but Statement II is correct.
(3) Both Statement I and Statement II are correct.
(4) Both Statement I and Statement II are incorrect.

SECTION - B

136. Select the **incorrect** statement.
(1) In grasses, the guard cells are dumb-bell shaped.
(2) The guard cells possess chloroplast.
(3) Subsidiary cells are absent in monocotyledonous leaf.
(4) Guard cells are bean shaped in dicotyledonous leaf.

137. In monocotyledonous stem:
(1) vascular bundles are scattered.
(2) vascular bundles are arranged in a ring and have cambium.
(3) xylem and phloem are radially arranged.
(4) Central vascular bundles are generally smaller than the peripherally located ones.

138. Select the **correct** match.
A. Respiratory substrate – Pure proteins
B. Energy currency of the cell – ATP
C. Gaseous exchange in plants– Lenticels
D. Yeast(fermentation)- Lactic acid
Choose the most appropriate answer from the options given below:
(1) B, C and D only
(2) A, B, C and D
(3) B and C only
(4) B and D only

139. The outer _____, consists of a few layers of collenchymatous cells just below the epidermis, which provide mechanical strength to the young stem in dicots.
(1) cortical layers
(2) hypodermis
(3) endodermis
(4) epiblema

140. All are incorrect about vascular bundles in monocotyledonous stem **except**:
(1) single vascular bundle is present.
(2) conjoint open vascular bundle.
(3) cambium present.
(4) surrounded by sclerenchymatous bundle sheaths.

141. Given below are two statements:

Statement I: During glycolysis, ATP is utilised in glycolysis.

Statement II: ATP is utilised during the conversion of phosphoenolpyruvate to pyruvic acid.

In the light of the above statements, choose the most appropriate answer from the options given below:

(1) Statement I is correct but Statement II is incorrect.
(2) Statement I is incorrect but Statement II is correct.
(3) Both Statement I and Statement II are correct.
(4) Both Statement I and Statement II are incorrect.

142. Identify the **incorrect** statement from the following.

(1) Trichomes prevent water loss.
(2) All tissues that are neither vascular nor epidermal tissues are called ground tissues.
(3) Medullary rays are parenchymatous.
(4) Pith is with very less intercellular spaces in dicotyledonous stem.

143. Hypodermis of a monocotyledonous stem is made of:

(1) sclerenchymatous cells.
(2) parenchymatous cells.
(3) collenchymatous cells.
(4) chlorenchymatous cells.

144. Trichomes are:

(1) epidermal hair of stem.
(2) either soft or stiff.
(3) branched or unbranched.
(4) All of these.

145. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:

Assertion A: The TCA cycle requires the continued replenishment of oxaloacetic acid.

Reason R: Citric acid is the last member of the TCA cycle.

In the light of the above statements, choose the **correct** answer from the options given below:

(1) A is true but R is false.
(2) A is false but R is true.
(3) Both A and R are true and R is the correct explanation of A.
(4) Both A and R are true but R is NOT the correct explanation of A.

146. Match **List-I** with **List-II**.

	List-I		**List-II**
(A)	ATP synthesis	(I)	Citrate to isocitrate
(B)	Reduction	(II)	NAD^+ to $NADH + H^+$
(C)	Decarboxylation	(III)	Release of CO_2 during alcohol fermentation
(D)	Isomerisation	(IV)	1,3-bisphosphoglycerate to 3-phosphoglyceric acid

Choose the **correct** answer from the options given below:

(1) A-II, B-III, C-I, D-IV
(2) A-IV, B-II, C-III, D-I
(3) A-II, B-III, C-IV, D-I
(4) A-II, B-I, C-III, D-IV

147. Which among the following is **not** a feature of monocotyledonous root?

(1) Epidermis
(2) Pith
(3) Pericycle
(4) Conjoint vascular bundles

148. The main purpose of electron transport chain is/are:

A. cycle $NADH + H^+$ back to NAD^+.
B. form acetyl CoA.
C. utilise glucose and sucrose.
D. use the intermediates from TCA cycle.

Choose the most appropriate answer from the options given below:

(1) B, C and D only
(2) B and C only
(3) A only
(4) A, B, C and D

149. There can be a net gain of _______ ATP molecules during aerobic respiration of one molecule of fructose.

(1) 34
(2) 2
(3) 5
(4) 38

150. The names of different tissues are given below:

Epidermis, hypodermis, endodermis, pericycle, vascular bundles, pith

How many of the above are present in stele?

(1) Three
(2) Four
(3) Five
(4) Six

SECTION-(IV) ZOOLOGY

SECTION - A

151. Select the **incorrect** statement.
(1) Neural system provides point-to-point rapid coordination among organs.
(2) Neural coordination is fast.
(3) Neural coordination is short-lived.
(4) Nerve fibres innervate all the body cells to regulate their functions continuously.

152. Choose the **incorrect** statement.
(1) The midbrain receives and integrates visual, tactile and auditory inputs.
(2) Exopthalmic goitre is a form of hyperthyroidism.
(3) Neural system of all animals is composed of highly specialised cells.
(4) Invertebrates have a more developed neural system.

153. Which of the following functions is influenced by the hormone of the pineal gland?
(1) Defense capability
(2) Body temperature regulation
(3) Menstrual cycle control
(4) All of these

154. Which of the following options is **correct** w.r.t thymosin?
(1) It is a peptide hormone.
(2) It is secreted by pituitary gland.
(3) It helps in RBCs production.
(4) It decreases WBCs production.

155. The hypothalamus is located in which part of the brain?
(1) Pons
(2) Midbrain
(3) Forebrain
(4) Medulla oblongata

156. Choose the **incorrect** statement.
(1) Neurons are excitable cells.
(2) The axons transmit nerve impulses towards the cell body.
(3) Visceral neural system is a part of PNS.
(4) Brain acts as a command and control centre.

157. Select the **incorrect** pair.

(1)	Ovary	Progesterone
(2)	α-cell of pancreas	Glucagon
(3)	Adrenal cortex	Mineralocorticoids
(4)	Follicular cells of parathyroid	TCT

158. The __X__ of kidney produces peptide hormone called __Y__ which stimulates erythropoiesis. Choose the options which fill the blanks **correctly**.

	X	Y
(1)	podocyte	erythropoietin
(2)	JG cells	erythropoietin
(3)	JG cells	relaxin
(4)	JG cells	renin

159. Select the **mismatch** option w.r.t types of neuron.

(1)	Multipolar	Cerebral cortex
(2)	Unipolar	Embryonic stage
(3)	Bipolar	Retina of eye
(4)	Myelinated nerve fibres	Commonly found in autonomic neural system

160. Receptors of neurotransmitters are present on:
(1) membrane of synaptic vesicle.
(2) pre-synaptic membrane.
(3) node of Ranvier.
(4) post-synaptic membrane.

161. Which of the following hormones directly affects transcription by acting on gene?
(1) FSH
(2) GH
(3) Estrogen
(4) CCK

162. Read the following statements **(A-E)** and select the **correct** statement.
A. Human brain is well protected by the skull.
B. Brain is the site for processing of vision, hearing, speech, memory, intelligence, emotions and thoughts.
C. The PNS comprises of all the nerves of the body associated with the CNS.
D. Cerebellum has very convoluted surface.
E. Arachnoid is in contact with the brain tissue.
Choose the *most appropriate* answer from the options given below:
(1) B, C, D and E only
(2) A, B and C only
(3) A, B, C and D only
(4) A, B, D and E only

163. Given below are two statements:

Statement I: The cerebral cortex is referred to as grey matter due to its greyish appearance.

Statement II: Resting axonal membrane is comparatively more permeable to potassium ions (K^+).

In the light of the above statements, choose the *most appropriate* answer from the options given below:

(1) Statement I is correct but Statement II is incorrect.
(2) Statement I is incorrect but Statement II is correct.
(3) Both Statement I and Statement II are correct.
(4) Both Statement I and Statement II are incorrect.

164. Choose the **correct** option.

(1) Unmyelinated nerve fibre is found in somatic neural systems.
(2) Only cell body of neurons contain Nissl's granules.
(3) Axon is a short fibre.
(4) Based on number of axons and dendrites, neurons are divided into 4 types.

165. Which of the following diseases is **not** linked to its respective hormone/gland?

(1)	Cretinism	Thyroid hormone
(2)	Addison's disease	Adrenal cortex hormone
(3)	Acromegaly	Growth hormone
(4)	Diabetes Insipidus	Insulin

166. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:

Assertion A: Insulin is a hypoglycemic hormone.

Reason R: Insulin stimulates glycogenesis.

In the light of the above statements, choose the **correct** answer from the options given below:

(1) A is true but R is false.
(2) A is false but R is true.
(3) Both A and R are true and R is the correct explanation of A.
(4) Both A and R are true but R is not the correct explanation of A.

167. Which of the following statements about the sympathetic nervous system is **correct**?

(1) It slows down the heart rate.
(2) It is a part of the central nervous system.
(3) It prepares the body for "fight or flight" responses.
(4) It is involved directly in digestion.

168. Somatic neural system relays impulses from:

(1) CNS to skeletal muscles.
(2) CNS to the smooth muscles.
(3) skeletal muscles to CNS.
(4) smooth muscles to CNS.

169. Match **List-I** with **List-II**:

List-I (Production Site)		**List-II (Hormones)**	
(A)	Atrial wall of heart	(I)	ANF
(B)	Thyroid gland	(II)	Stimulates secretion of pancreatic enzyme
(C)	CCK	(III)	T_3
(D)	GIP	(IV)	Inhibits gastric secretion

Choose the **correct** answer from the options given below:

(1) A-II, B-IV, C-I, D-III
(2) A-I, B-III, C-II, D-IV
(3) A-II, B-IV, C-III, D-I
(4) A-IV, B-III, C-II, D-I

170. A. Anabolic effect on protein and carbohydrate metabolism.

B. Influences male sexual behavior (libido).

C. Stimulates spermatogenesis, aggressiveness and low pitch voice.

Above are the functions of which of the following hormones?

(1) Estrogens
(2) Progesterone
(3) Testosterone
(4) Relaxin

171. Read the following statements **(A-D)** w.r.t. pituitary gland.

A. The pituitary gland is located in a bony cavity called sella tursica.

B. It is divided anatomically into adenohypophysis and neurohypophysis.

C. Pars intermedia produces adrenocortico-trophic hormone.

D. The anterior pituitary is under the direct neural regulation of the hypothalamus.

Choose the *most appropriate* answer from the options given below:

(1) Only A and D are correct.
(2) Only C and D are correct.
(3) Only A, B and D are correct.
(4) Only A and B are correct.

172. The dorsal portion of the midbrain consists:
(1) corpora quadrigemina.
(2) pons.
(3) cerebellum.
(4) medulla.

173. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:
Assertion A: The nerve fibres of the PNS are of two types.
Reason R: Afferent fibers send impulses from tissues to the CNS, while efferent fibers relay signals from the CNS to tissues.
In the light of the above statements, choose the **correct** answer from the options given below:
(1) A is true but R is false.
(2) A is false but R is true.
(3) Both A and R are true and R is the correct explanation of A.
(4) Both A and R are true but R is not the correct explanation of A.

174. Which of the following is a **incorrect** match?

(1)	Zona fasciculata	Middle layer of adrenal cortex
(2)	Glucocorticoids	Stimulate lipolysis, RBC production
(3)	T-lymphocytes	Humoral immunity
(4)	PTH	Increases the Ca^{++} levels in blood

175. Reabsorption of water and electrolytes by distal tubules of kidney is done by:
(1) oxytocin. (2) vasopressin.
(3) FSH. (4) LH.

176. Non-endocrine tissues secrete several hormones called:
(1) growth factors.
(2) inhibiting hormones.
(3) releasing hormones.
(4) gonadotrophins.

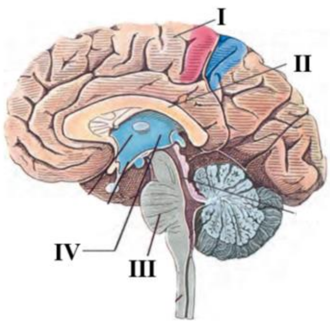
177. Given below are two statements:
Statement I: Cyclic AMP is a second messenger.
Statement II: Estrogen is synthesised and secreted mainly by the growing ovarian follicles.
In the light of the above statements, choose the *most appropriate* answer from the options given below:
(1) Statement I is correct but Statement II is incorrect.
(2) Statement I is incorrect but Statement II is correct.
(3) Both Statement I and Statement II are correct.
(4) Both Statement I and Statement II are incorrect.

178. Visceral nervous system comprises of:
(1) plexuses only.
(2) ganglia only.
(3) nerves only.
(4) all of these

179. Sodium-potassium pump transports:
(1) 3 Na^+ outwards for 2 K^+ into the cell.
(2) 2 Na^+ outwards for $3K^+$ into the cell.
(3) 2 Na^+ outwards for $2K^+$ into the cell.
(4) 2 Na^+ and $3K^+$ outwards to the cell.

180. Which hormone is produced by the pars distalis to regulate the growth of mammary glands and milk production?
(1) Growth hormone (GH)
(2) Thyroid stimulating hormone (TSH)
(3) Prolactin (PRL)
(4) Follicle stimulating hormone (FSH)

181. Identify different parts of the brain given below in the diagram and mark the option that shows **correct** identification of characteristic of part labelled as **I**, **II**, **III**, **IV**.



(1) I - Has inner cortex and outer medulla
(2) II - Connects the two hemisphere
(3) III - Control body temperature
(4) IV - Regulate complex functions like memory and communication

182. The parts of human brain that helps in regulation of sexual behaviour, expression of excitement, pleasure, rage, fear etc. are:
(1) limbic system and hypothalamus.
(2) corpus callosum and thalamus.
(3) brainstem and epithalamus.
(4) corpora quadrigemina and hippocampus.

183. The pars distalis of the pituitary secretes all of the following hormones, **except**:
(1) follicle-stimulating hormone.
(2) melanocyte stimulating hormone.
(3) growth hormone.
(4) prolactin.

184. Secretion of progesterone by corpus luteum is initiated by:

(1) testosterone.

(2) thyroxine.

(3) MSH.

(4) LH.

185. Match **List-I** with **List-II:**

List-I		List-II	
(A)	Pons	(I)	Provides additional space for neurons
(B)	Hypothalamus	(II)	Controls respiration and gastric secretions
(C)	Medulla	(III)	Connects different regions of the brain
(D)	Cerebellum	(IV)	Neuro-secretory cells

Choose the **correct** answer from the options given below:

(1) A-III, B-I, C-IV, D-II

(2) A-II, B-I, C-IV, D-III

(3) A-III, B-IV, C-II, D-I

(4) A-IV, B-I, C-III, D-II

SECTION - B

186. Brainstem of human brain consists of:

(1) mid brain, pons, medulla oblongata.

(2) forebrain, cerebellum and pons.

(3) thalamus and hypothalamus only.

(4) amygdala and hippocampus only.

187. Select the **incorrect** statement regarding synapses.

(1) In an electrical synapse, the membranes of presynaptic and postsynaptic neurons are closely aligned.

(2) Electrical current flows directly from one neuron to another in electrical synapses.

(3) Electrical synapses are rare in our system.

(4) Impulse transmission in a chemical synapse is always faster than in an electrical synapse.

188. During nerve impulse transmission, the inner membrane potential changes as follows:

(1) initially positive, then negative, and remains negative.

(2) initially negative, then positive, and remains positive.

(3) initially positive, then negative, and returns to positive.

(4) initially negative, then positive, and returns to negative.

189. In human females ___X___ stimulates a vigorous contraction of uterus at the time of child birth.
Choose the option which fills the blank **correctly**.

(1) LH (2) FSH

(3) oxytocin (4) relaxin

190. Which of the following is the major coordinating centre for sensory and motor signaling in the brain?

(1) Hypothalamus

(2) Thalamus

(3) Corpora quadrigemina

(4) Amygdala

191. Match **List-I** with **List-II:**

List-I		List-II	
(A)	Medulla	(I)	Olfaction
(B)	Association areas	(II)	Memory
(C)	Limbic system	(III)	Concentration of cell bodies
(D)	Grey matter	(IV)	Gastric secretions

Choose the **correct** answer from the options given below:

(1) A-IV, B-I, C-II, D-III

(2) A-I, B-IV, C-III, D-II

(3) A-III, B-I, C-IV, D-II

(4) A-IV, B-II, C-I, D-III

192. How many from the following is/are secreted by pituitary gland?

TSH, GH, PRL, ACTH

(1) One

(2) Two

(3) Four

(4) Three

193. Autonomic neural system transmits impulses from CNS to:

(1) involuntary organs.

(2) smooth muscles.

(3) skeletal muscles.

(4) both (1) and (2)

194. Given below are two statements:
Statement I: Oxytocin causes milk ejection from mammary gland.
Statement II: Catecholamines increase the breakdown of glycogen and lipids.
In the light of the above statements, choose the *most appropriate* answer from the options given below:

(1) Statement I is correct but Statement II is incorrect.

(2) Statement I is incorrect but Statement II is correct.

(3) Both Statement I and Statement II are correct.

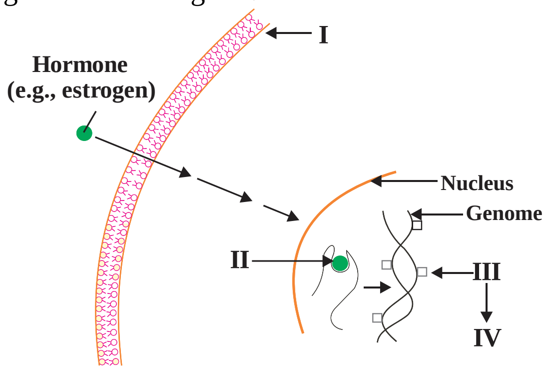
(4) Both Statement I and Statement II are incorrect.

195. Depolarisation of neural membrane is caused due to:
(1) rapid influx of K^+ ions.
(2) rapid efflux of Na^+ ions.
(3) rapid influx of Cl^- ions.
(4) rapid influx of Na^+ ions.

196. Source of somatostatin is the same as that of:
(1) insulin and glucagon.
(2) vasopressin and oxytocin.
(3) thyroxine and calcitonin.
(4) somatotropin and prolactin.

197. Choose the **correct** statement w.r.t myelinated nerve fibres.
(1) Nodes of Ranvier are absent in them.
(2) Enveloped by schwann cells.
(3) Schwann cells neither enclose nor envelop them.
(4) Found only in spinal nerves.

198. Identify the **incorrect** labelling out of **I** to **IV** in the given below diagram.



(1) I - Uterine cell membrane.
(2) II - Receptor-hormone complex.
(3) III - tRNA.
(4) IV - Physiological response

199. Read the following statements **(A-E)**.
A. The thymus gland is a lobular structure.
B. Aldosterone acts mainly at the renal tubules and stimulates the reabsorption of Cl^-.
C. Leydig cells produce a group of hormones called androgens.
D. Thymus is degenerated in old individuals.
E. Insulin deficiency leads to increased glucose uptake by cells.

Choose the *most appropriate* answer from the options given below:
(1) Only A and E are correct.
(2) Only C and D are correct.
(3) Only A, C and D are correct.
(4) Only A, B, D and E are correct.

200. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:
Assertion A: Posterior pituitary is also known as pars nervosa.
Reason R: Posterior pituitary stores and releases three hormones.
In the light of the above statements, choose the **correct** answer from the options given below:
(1) A is true but R is false.
(2) A is false but R is true.
(3) Both A and R are true and R is the correct explanation of A.
(4) Both A and R are true but R is not the correct explanation of A.